SHUBHAM JOSHI

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Live agent platform | Product demo | Live model | Portfolio | github.com/shubhamjoshipromail-svg | linkedin.com/in/shubham-joshi1

SUMMARY

AI engineer who builds intelligent applications on top of foundation models, ships them to real users, then makes them reliable enough to keep running. I design the agent behaviour, write the backend, build the interface, fine tune the model when the model is the problem, and write the evaluation that decides whether it works. In the last year: 508 commits and 92,448 lines of code across 11 public repositories, 8 deployed and live right now, a fine tuned transformer on Hugging Face, and 40 technical specifications totaling 6,700 lines. I build the whole thing, then I try to break it, and I report what broke.

SKILLS

Agentic AI, LLMs and Emerging Technologies: Agentic AI systems and agent architectures; Large Language Models and foundation models (Anthropic Claude, OpenAI GPT and Codex, Google Gemini); designing agent behaviour end to end: prompt engineering, reasoning and planning, tool and function calling, multi agent orchestration, Model Context Protocol servers and clients, retrieval and context engineering, structured output, multi provider routing and cost control; AI powered development with Cursor, Claude Code and Codex; familiarity with the frontier of applied AI, developing and contributing reusable tooling to the broader open source ecosystem

Evaluation, Guardrails and AI Safety: Evaluation harnesses and benchmark design, prespecified protocols with frozen fixtures, scenario and regression testing, adversarial and red team testing including prompt injection defense, deny by default permissions, human in the loop approval gates, secure credential handling, audit trails, confidence gating and abstention, honest reporting of negative results, responsible AI

Machine Learning and Deep Learning: PyTorch, Transformers and Hugging Face, transformer fine tuning and public model deployment (ModernBERT), scikit-learn, LightGBM, XGBoost, leakage safe splits, calibration, drift monitoring, SHAP and LIME interpretability, error analysis, taxonomy and label design, model selection under cost and latency budgets

Applied Computer Science and Software Engineering: Applied computer science and programming: Python, TypeScript, JavaScript, SQL; Next.js, React, Node, FastAPI, PostgreSQL, Prisma, REST APIs, system design, streaming over server sent events, durable job queues, idempotency, crash recovery, concurrency control, observability, performance profiling, cost tracking, automated testing, Docker, Git, CI/CD, services deployed and operated in production on Railway, Chrome extension and Tauri packaging; comfortable embedded in an unfamiliar codebase picking up new tasks

Collaboration and Communication: Cross-functional collaboration: collaborate with engineers, designers and non technical stakeholders, partner directly with clients to identify problems worth solving and develop integrated solutions that drive measurable impact, contribute to shared tooling and support colleagues, strong written and spoken communication skills, technical documentation, presenting findings, teaching and mentorship, continuous improvement

PROJECTS

AgentDock, Agent Platform That Takes Real Actions Live app | Demo | Pitch deck | Code

- Designed how the agent thinks and built all of it solo: model backed planning compiling a plain language goal into a validated multi step workflow, a reasoning and tool call layer executing over the real Model Context Protocol wire, and four first party connectors for Gmail, Calendar, Docs and Search. Rather than hardcode a tool list it syncs and classifies the official MCP registry by category, risk and verification, so the catalog grows with the ecosystem, not with my releases.
- Solved what actually blocks agents from touching a real account, which is trust rather than capability: permissions resolve and clamp server side so prompt text can request capability but never grant it, every outbound action passes an approval gate, credentials are encrypted and budgets capped, and I wrote the adversarial suite proving prompt injection through tool output cannot fire an unapproved send.
- Engineered it for production: 225 commits and 35,539 lines in two months with 11,000 lines of tests across 66 test files, 33 API routes, 26 migrations, a PostgreSQL job queue with lease and heartbeat crash recovery, idempotent external actions, concurrency caps, live run streaming, and 24 design documents written before the code.

Vibe Curator, Generative AI Product in Three Form Factors Live app | Code and releases

- Built a real time creative environment across 104 commits and 19,913 lines where four providers land in one editable scene model, so a generation is starting material rather than a flattened file: OpenAI for images, Anthropic for shaders, Gemini Veo for video, ElevenLabs for music. It ships three ways at once: a live web app, a Chrome extension with Web Store release assets, and a signed Tauri desktop build for macOS.
- Made unreliable generation dependable by treating the model as an untrusted code author, the general lesson for any code generation feature: model written GLSL clears a static guard and a real compile pass before it renders, and line numbered compiler errors feed back so repairs converge within 3 attempts.

OpsPilot and RxCheck, Fine Tuned and Evaluated AI Live model | Live app | Research report

- Fine tuned and published a ModernBERT ticket router on 23,747 leakage safe examples, diagnosing instead of scaling: baselines put TF-IDF at 44 and ModernBERT at 52.3 percent, so the labels were the ceiling, and rebuilding ten labels into seven queues lifted accuracy to 73.1 percent, shipped with abstention that auto routes 62.3 percent at 85.5 percent accuracy.
- RxCheck reviews 152,416 records where the database decides and the model only explains, passing 26 of 26 scenarios even with every external service failing, and its research report is where I published the results that made me look bad: my own validator accepted 15 malformed outputs of 30, and an official external benchmark passed only 22 of 30.

EXPERIENCE

Analytics Consultant, Johns Hopkins University, Client: Alliance of Community Health Plans - Washington, DC Jul 2025 - Jul 2026

- Led technical execution for a six person team on a national client engagement, building the Python and SQL pipeline that reconciled four inconsistent federal sources into a 138,263 row dataset, catching a twelvefold enrollment overcount before it reached a model, and presenting in person to leadership, who said it would change where the organization spent its resources.

Co-Founder and Vice President, Business Analytics and AI Organization, Johns Hopkins University - Washington, DC Jan 2026 - Present

- Co-founded and lead a 100 member organization alongside my degree, delivering a 3 hour hands on workshop on agents and modern AI development to 35 registrants and publishing the guide open source.

Founder, Digital Marketing and Web Studio, self-employed - Nagoya, Japan Jan 2021 - May 2025

- Ran a studio single handed for four years for international clients in a third language, owning scoping, pricing, delivery and reporting.

EDUCATION

Johns Hopkins University, Carey Business School - MS, Business Analytics and Artificial Intelligence, STEM designated, Dean's Scholarship Aug 2026

Nagoya University of Commerce and Business - BBA in Management, **GPA 3.63** (WES evaluated), Dean's Scholar honors, Monash exchange 2025

Coursework: Generative AI, Practical Machine Learning, Sequential Decision Making, Responsible AI, Statistical Analysis. **Certifications:** Anthropic Claude 101, Anthropic AI Fluency, Azure AI-900. **Languages:** English, Japanese, Nepali, Hindi.